



KCED
Learning Augmented

24ADI003-MACHINE LEARNING

Course Objectives:

The purpose of taking this course is to:

1. Introduce the fundamental concepts of machine learning, its life cycle and ethical considerations.
2. Explore various supervised and unsupervised learning techniques and optimization strategies.
3. Examine recommendation systems, and its evaluation techniques.

Course Outcomes

- After successful completion of this course, the students shall be able to
- **CO1** Apply appropriate **data pre-processing techniques** to build machine learning models with ethical considerations. (Ap)
- **CO2** Analyze and optimize **regression models using estimation techniques, regularization, and gradient-based methods** through error analysis. (An)
- **CO3** Build and evaluate the effectiveness of **different classification models and ensemble techniques**. (E)
- **CO4** Analyze complex datasets using advanced **clustering, associative rule mining and dimensionality reduction algorithms** to uncover meaningful patterns and groupings. (An)
- **CO5** Develop **recommendation systems** to personalize user needs. (Ap)

Course Content

- **INTRODUCTION**

Introduction to Machine Learning-Types of machine learning: Supervised, Unsupervised, Semi supervised and Reinforcement Learning-Applications of machine learning in various fields- Ethics in machine learning- Fairness, accountability and interpretability -Machine learning workflow- Data Preprocessing-Feature engineering- Correlation analysis- Model training and evaluation- Model monitoring and maintenance.

- **REGRESSION MODELS**

Linear regression- Simple Regression-Least Square Estimator-Maximum Likelihood Estimator- Multiple Regression -Polynomial Regression-Performance Metrics-Bias

- **CLASSIFICATION MODELS**

Logistic Regression-Naive Bayes Classifiers-Decision Tree-K-Nearest Neighbors-Support Vector Machine – Evaluation metrics – AUC ROC- Class Imbalance –SMOTE – Cross-Validation Techniques-Ensemble Learning- Bagging- Random Forests-Boosting - AdaBoost -Gradient Boosting

- **UNSUPERVISED LEARNING**

Clustering- K-means Clustering- Gaussian Mixture Models -Hierarchical Clustering- Density-Based Clustering (DBSCAN)- Mean-Shift Clustering- Spectral Clustering- Association Rule Learning- Apriori Algorithm- FP-Growth Algorithm- Dimensionality Reduction- Principal Component Analysis (PCA)- Linear Discriminant Analysis (LDA)

- **RECOMMENDATION SYSTEMS**

Introduction to Recommendation Systems-Types- Challenges- Collaborative Filtering Techniques- User-Based- Collaborative Filtering- Item-Based- Collaborative Filtering- Matrix Factorization Techniques- Content-Based Recommendation- Hybrid Recommendation Systems- Evaluation of Recommendation Systems

Unit III – CLASSIFICATION MODELS

Logistic Regression-Naive Bayes Classifiers-Decision Tree-K-Nearest Neighbors-Support Vector Machine – Evaluation metrics – AUC ROC- Class Imbalance – SMOTE – Cross-Validation Techniques-Ensemble Learning-Bagging- Random Forests-Boosting - AdaBoost -Gradient Boosting

NAIVE BAYES CLASSIFIER

In many real-world problems, we must **classify objects**:

- Is an email **Spam or Not Spam**?
- Will a student **Pass or Fail**?
- Is a transaction **Fraud or Genuine**?

Naive Bayes solves **classification problems** by using **probability theory**.

Instead of guessing directly, it asks:

“Given what I see, which class is most likely?”

Bayes' Theorem

- Bayes' theorem allows us to **reverse conditional probabilities**:

$$P(C | X) = \frac{P(X | C) P(C)}{P(X)}$$

Term	Meaning
$P(C)$	Prior probability (how frequent the class is)
$P(X)$	Evidence (same for all classes)

Since $P(X)$ is the same for all classes:

$$P(C | X) \propto P(X | C) P(C)$$

→ We **do NOT calculate** $P(X)$ in practice.

NAIVE BAYES CLASSIFIER

Why “Naive”?

Naive Bayes assumes:

Features are conditionally independent given the class

- Mathematically:

$$P(X_1, X_2, \dots, X_n | C) = P(X_1 | C) \times P(X_2 | C) \times \dots \times P(X_n | C)$$

- If the class is known:
- Knowing feature 1 gives **no information** about feature 2

Example:

- If an email is spam:
 - Word “Free” does not affect word “Win”
- This assumption is **naive** because in reality:
- Words are correlated
- Yet:
Model works well in practice
Especially with large data

NAIVE BAYES CLASSIFIER

Example (Spam Classification)

- Email contains:

“Free Win Offer”

- Naive Bayes treats it as:

$$P(\text{Free, Win, Offer} | \text{Spam}) = P(\text{Free} | \text{Spam}) P(\text{Win} | \text{Spam}) P(\text{Offer} | \text{Spam})$$

NAIVE BAYES CLASSIFIER

Types of Naive Bayes

1 Gaussian Naive Bayes

- Used when:
- Features are **continuous**
- Assumes **normal distribution**
- Example:
- Height
- Marks
- Temperature

2 Multinomial Naive Bayes

- Used when:
- Features are **counts**
- Example:
- Word frequency in documents

3 Bernoulli Naive Bayes

Used when:

Features are **binary (0/1)**

Example:

Word present or absent

NAIVE BAYES CLASSIFIER

Problem Statement 1

- Classify an email as **Spam or Ham**
- Given:

Word	P(Word Spam)	P(Word Ham)
Free	0.6	0.1
Offer	0.5	0.2

- Prior probabilities:

$$P(\text{Spam}) = 0.4$$

$$P(\text{Ham}) = 0.6$$

- Email text:

“Free Offer”

Step 1: Calculate Spam Probability

$$P(\text{Spam} | X) \propto P(\text{Spam}) \times P(\text{Free} | \text{Spam}) \times P(\text{Offer} | \text{Spam})$$

$$= 0.4 \times 0.6 \times 0.5 = 0.12$$

NAIVE BAYES CLASSIFIER

Problem Statement 1

- Classify an email as **Spam or Ham**
- Given:

Word	P(Word Spam)	P(Word Ham)
Free	0.6	0.1
Offer	0.5	0.2

- Prior probabilities:
 $P(\text{Spam}) = 0.4$
 $P(\text{Ham}) = 0.6$
- Email text:

“Free Offer”

Step 2: Calculate Ham Probability

$$P(\text{Ham} | X) \propto 0.6 \times 0.1 \times 0.2 = 0.012$$

Step 3: Compare Scores

Class	Score
Spam	0.12
Ham	0.012

Prediction: Spam

NAIVE BAYES CLASSIFIER

Problem Statement 2

- Given Data

Word	P(Word Spam)	P(Word Ham)
Free	0.7	0.1
Win	0.6	0.05
Hello	0.1	0.6

- Priors:

$$P(\text{Spam}) = 0.3$$

$$P(\text{Ham}) = 0.7$$

- Email:

- “Free Win”

Spam Score

$$0.3 \times 0.7 \times 0.6 = 0.126$$

Ham Score

$$0.7 \times 0.1 \times 0.05 = 0.0035$$

→ **Spam detected**

NAIVE BAYES CLASSIFIER

- **Problem Statement** (Gaussian Naive Bayes)
- We want to **predict whether a student Passes or Fails** based on their **marks** using **Gaussian Naive Bayes**.
- Given Training Statistics

Class	Mean (μ)	Standard Deviation (σ)
Pass	70	10
Fail	40	8

- **New Student**

Marks $x = 65$

- Assume **equal priors**:

$$P(\text{Pass}) = P(\text{Fail})$$

- So decision depends **only on likelihoods**.

Step 1: Gaussian Probability Density Function

Gaussian Naive Bayes assumes:

$$P(x | C) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

NAIVE BAYES CLASSIFIER

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Step 1: Gaussian Probability Density Function

Gaussian Naive Bayes assumes:

$$P(x | C) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

Step 2: Likelihood for PASS Class Substitute Values

$$\mu_{pass} = 70, \sigma_{pass} = 10, x = 65$$

Exponent Term

$$(x - \mu)^2 = (65 - 70)^2 = 25$$

$$2\sigma^2 = 2 \times 10^2 = 200$$

$$-\frac{25}{200} = -0.125$$

$$e^{-0.125} \approx 0.8825$$

Normalization Term

$$\sqrt{2\pi\sigma^2} = \sqrt{2\pi \times 100} \approx 25.07$$

Final Likelihood

$$P(65 | Pass) = \frac{1}{25.07} \times 0.8825 \approx 0.0352$$

NAIVE BAYES CLASSIFIER

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- We want to **predict whether a student Passes or Fails** based on their **marks** using **Gaussian Naive Bayes**.
- Given Training Statistics

Class	Mean (μ)	Standard Deviation (σ)
Pass	70	10
Fail	40	8

Step 1: Gaussian Probability Density Function

Gaussian Naive Bayes assumes:

$$P(x | C) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

Step 3: Likelihood for FAIL Class Substitute Values

$$\mu_{Fail} = 40, \sigma_{Fail} = 8$$

Exponent Term

$$(x - \mu)^2 = (65 - 40)^2 = 625$$

$$2\sigma^2 = 2 \times 8^2 = 128$$

$$-\frac{625}{128} \approx -4.88$$

$$e^{-4.88} \approx 0.0076$$

Normalization Term

$$\sqrt{2\pi\sigma^2} = \sqrt{2\pi \times 64} \approx 20.05$$

Final Likelihood

$$P(65 | Fail) = \frac{1}{20.05} \times 0.0076 \approx 0.00038$$

NAIVE BAYES CLASSIFIER

- **Problem Statement** (Gaussian Naive Bayes)
- We want to **predict whether a student Passes or Fails** based on their **marks** using **Gaussian Naive Bayes**.
- Given Training Statistics

Class	Mean (μ)	Standard Deviation (σ)
Pass	70	10
Fail	40	8

Step 4: Compare Likelihoods

Class	Likelihood
Pass	0.0352
Fail	0.00038

$$P(65|Pass) \gg P(65|Fail)$$

Step 5: Final Prediction

Since priors are equal:

$$P(Pass | 65) > P(Fail | 65)$$

Predicted Class: PASS

- 65 is **close to the Pass mean (70)**
- 65 is **very far from the Fail mean (40)**
- Gaussian curve assigns:
- High probability near the mean
- Very low probability far from the mean

NAIVE BAYES CLASSIFIER

- Handling Zero Probability – Laplace Smoothing (Add-One Smoothing)

1. The Zero Probability Problem (Why It Is Serious)

- In Naive Bayes, we compute:

$$P(C | X) \propto P(C) \prod_i P(X_i | C)$$

- This means:
- We **multiply probabilities** of all features

What Goes Wrong?

- If **any one feature** has:

$$P(X_i | C) = 0$$

- Then:
- Entire *product* = 0
- Even if all other features strongly support the class
Model **completely rejects** that class

NAIVE BAYES CLASSIFIER

- Handling Zero Probability – Laplace Smoothing (Add-One Smoothing)

2. Why Zero Probability Occurs

- Zero probability happens when:
- A word **never appeared** in training data for a class

Example:

- Word “Lottery” never appeared in **Ham emails**
- Then:

$$P(\text{Lottery} | \text{Ham}) = 0$$

Just because a word **never appeared before**, it does **not mean it can never appear**.

- Training data is limited
 - Language evolves
 - Rare events exist
- Zero probability is **too harsh** and unrealistic

NAIVE BAYES CLASSIFIER

- Handling Zero Probability – Laplace Smoothing (Add-One Smoothing)

4. Laplace Smoothing – The Solution

- Laplace smoothing **adds 1** to every count to avoid zeros.
- **Formula**

$$P(\text{word} \mid C) = \frac{\text{count}(\text{word}, C) + 1}{\text{total words in } C + \text{vocabulary size}}$$

NAIVE BAYES CLASSIFIER

- **Example**
- **Training Data**
- Spam emails:
 - Total words = 100
 - Vocabulary size = 20
- Word “Free” appears:
 - 5 times in Spam
- Word “Lottery” appears:
 - 0 times in Spam

Without Laplace Smoothing

$$P(\text{Lottery}/\text{Spam}) = \frac{0}{100} = 0$$

- Entire Spam probability becomes zero

With Laplace Smoothing

$$P(\text{Lottery}/\text{Spam}) = \frac{0 + 1}{100 + 20} = \frac{1}{120}$$

- Probability is small but **non-zero**
Model remains stable